Beyond ELAN

Toward Open, Modular Infrastructure for Multimodal Annotation

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ELAN has been the de facto standard for multimodal language annotation for over two decades. It was designed, however, for a fundamentally different research context: small corpora, entirely manual workflows, desktop-local data, and no expectation of cross-institutional sharing. Today, the field operates at a scale and with requirements—large video corpora, AI-assisted analysis, GDPR compliance, real-time collaboration—that ELAN's monolithic architecture cannot accommodate. We survey the current annotation tool landscape, identify three structural gaps, and argue that the path forward is not a new ELAN but an *open annotation stack*: a community-governed, modular infrastructure in which privacy, AI integration, and interoperability are first-class design primitives.

Introduction

Multimodal language research—spanning gesture studies, sign language linguistics, conversation analysis, and child language acquisition—has long depended on video as its primary data source. For most of the past two decades, annotating that video has meant one thing: ELAN.

Developed at the Max Planck Institute for Psycholinguistics in the late 1990s, ELAN gave researchers a structured way to time-align annotations across multiple tiers (Wittenburg et al. 2006). It became indispensable not because it was the best imaginable tool but because it was the first tool good enough to become shared infrastructure. Researchers built workflows around it, trained graduate students on it, and deposited EAF files in discipline archives. ELAN is, in the deepest sense, path-dependent. [1]

The problem is that the world around ELAN has changed substantially, and ELAN has not. Three transformations in particular have opened a gap between what the tool provides and what researchers now need:

1. **Scale.** Corpora that once comprised dozens of hours of video now routinely comprise thousands. Manual annotation at that scale is economically and temporally untenable without AI assistance.

2. **AI.** Automated transcription, body-pose estimation, and gesture segmentation have matured from research prototypes to deployable components. ELAN provides no interface for them.

[1] The ELAN Annotation Format (EAF) is an XML schema that encodes tiers, time slots, and media linkages. Its stability as a format—the same files round-trip across two decades of ELAN versions—has been a genuine asset. It is the format's surrounding tool ecosystem, not the format itself, that has stagnated.

3. **Privacy and regulation.** GDPR and its equivalents have made the unprocessed sharing of research video legally fraught. Researchers either forgo sharing entirely or assemble ad-hoc anonymization scripts upstream of ELAN. Privacy remains structurally outside the tool.

These are not feature requests. They are structural mismatches between a tool's design assumptions and the field's current reality. Incremental patches to ELAN will not close them.

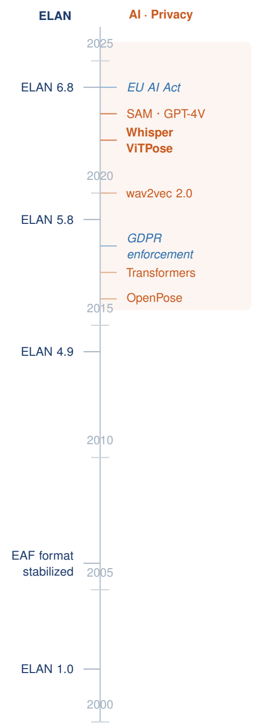

Figure 1: ELAN releases (left) have been sparse and incremental. AI capabilities (orange) and privacy regulation (blue) have accelerated sharply since 2016. The divergence is structural, not incidental.

The Annotation Landscape: Three Structural Gaps

Closed, Monolithic Architecture

ELAN is desktop software. It reads and writes EAF—an XML schema with no query API, no streaming interface, and no mechanism for external processes to read or modify annotation state during a session. Integrations are possible only by reading and writing EAF files on disk, a pattern that treats ELAN as a file format rather than a platform. [2]

Modern research pipelines are compositional. A researcher might use Whisper for speech transcription, ViTPose for skeletal tracking, a custom segmentation model for gesture boundaries, and ELAN for manual correction. Each step should produce outputs that feed cleanly into the next. EAF-on-disk integration makes this possible in principle but brittle in practice: format changes break parsers, file locking causes conflicts, and there is no shared notion of provenance. The tool that standardized annotation practice has standardized a *format* while leaving the surrounding workflow fragmented.

[2] A small ecosystem of conversion utilities exists—`pympi`, `elan-parser`, various lab-internal scripts—but each is maintained by a different group, none share a common abstraction layer, and all are fragile to ELAN version changes.

This is not merely a developer inconvenience. It means that the choices made by one lab—which parser, which conversion script, which naming convention—cannot be shared with another. Every lab that wants to add an AI component to its ELAN workflow must reinvent the integration from scratch. The cumulative cost to the field is substantial.

No AI Integration Path

The past five years have produced a set of AI capabilities directly relevant to multimodal annotation. Whisper (Radford et al. 2023) transcribes speech across dozens of languages with accuracy rivaling human transcription on clean audio. ViTPose (Xu et al. 2022) and its successors estimate full-body pose from video at real-time speed. Gesture segmentation models, gaze estimators, and emotion classifiers now exist as open-source releases. These tools do not eliminate the need for annotation; they change its character.

Human annotators are no longer asked to produce first-pass transcriptions from silence, but to correct, adjudicate, and interpret outputs that machines produce cheaply. The bottleneck shifts

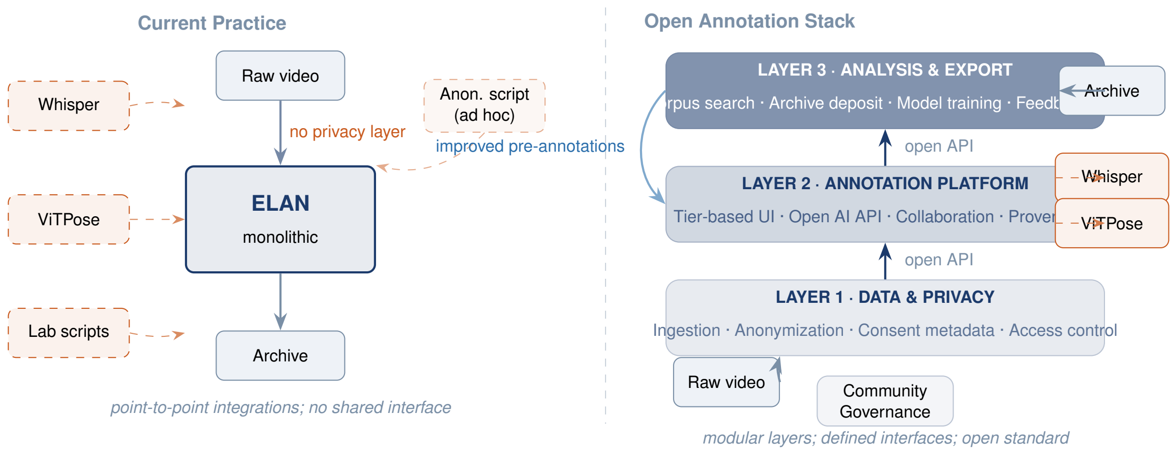

Figure 2: **Left:** The current ELAN-centric landscape, in which AI tools and anonymization scripts are ad-hoc external additions with no principled interface. Privacy is upstream and invisible to the annotation layer. **Right:** The proposed open annotation stack, organized into three layers with defined interfaces. Privacy is Layer 1; AI tools plug into Layer 2 via an open API; archival and analytical consumers connect to Layer 3.

from transcription speed to *interface quality*: how does a researcher efficiently review and correct AI outputs within a structured, tier-based environment?

ELAN has no answer. The assumption that annotators produce all content is baked into the tool's architecture. Researchers who want AI pre-annotation must run their models outside ELAN, convert outputs to EAF, and import them—a process that is slow, error-prone, and difficult to audit. [3] The field is replicating engineering effort at scale.

[3] Several research groups have independently built ELAN importers for Whisper output. None are mutually compatible; none track the distinction between machine-generated and human-corrected tiers as first-class metadata; none generalize to other models.

Privacy as Afterthought

ELAN operates on raw video. It annotates the recording as captured, with faces, voices, and incidentally identifying details fully present. This was unproblematic when research video stayed on a single lab workstation. It is increasingly untenable under GDPR, institutional data governance requirements, and the expectations of ethics review boards.

Current practice treats anonymization as a preprocessing step: blur faces and alter voices before the data enters the annotation workflow. This is better than nothing, but it externalizes privacy from the research infrastructure. Researchers who want to share both anonymized media and accompanying annotations face the additional problem that anonymization transforms may invalidate annotations: a blurred face cannot be coded for facial expression; a pitch-shifted voice cannot be analyzed for prosody. [4]

[4] This is not a hypothetical problem. Cross-institutional data-sharing initiatives in sign language research and multimodal dialogue corpora regularly stall on exactly this issue: the anonymization needed to share data is incompatible with the annotation resolution needed to analyze it.

What is missing is a privacy layer *integrated with* annotation: one that tracks the relationship between raw and de-identified media, allows annotations to reference identifiable content in a controlled-access environment, and treats consent and governance as first-class metadata rather than a note in a data management plan.

The Hidden Corpus: EAF Files in the Wild

The argument for new infrastructure is strengthened by an underappreciated fact: the old infrastructure has already produced a large and valuable corpus. ELAN has been in active use since the early 2000s. In the intervening two decades, researchers across linguistics, sign language studies, gesture research, conversation analysis, and child language acquisition have produced EAF files at scale. Major archival repositories—the Max Planck Institute's The Language Archive (TLA), the Endangered Languages Archive (ELAR) in London, PARADISEC in Australia, AILLA at the University of Texas, and the DOBES programme—each hold thousands of recorded sessions, many annotated in EAF.[5]

Beyond formal archives, the distribution of EAF files is diffuse and largely invisible. Every lab that uses ELAN has accumulated annotation files over years or decades of research. Field linguists returning from documentation trips have EAF files on laptops. Graduate students who annotated corpora for dissertations have EAF files on university servers. Many of these files are not archived; many that are archived cannot be shared because the accompanying video remains identifiable. The annotation labor they encode—if average annotation runs at a 10:1 ratio (ten hours of annotation per hour of video), a conservative estimate across known archives alone runs to hundreds of thousands of person-hours—is effectively locked.

[5] The DOBES programme (Documentation of Endangered Languages) ran from 2000 to 2015 and produced over 2,500 hours of field recordings across 60 language projects (VW-Stiftung 2015). TLA currently holds over 15,000 hours of annotated media. These are only the publicly archived fraction.

This is not merely a data management problem. It is a structural consequence of the same gap identified in Section : because there is no privacy layer between raw video and annotation, the video and the annotation are bound together in a way that makes sharing either one contingent on controlling the other. The EAF file without its video is incomplete; the video without anonymization is ungovernable.

Masking@Home: Unlocking the Heritage Corpus

We propose a community-driven mechanism for unlocking this heritage corpus, drawing on the tradition of citizen science distributed computation (Anderson et al. 2002; Lintott et al. 2008). *Masking@Home* would operate as an open platform through which holders of EAF-annotated video corpora can submit their materials for community-assisted anonymization, and through which verified contributors—research groups, community members, trained volunteers—process video through Layer 1 anonymization pipelines and return de-identified media alongside the original EAF annotations.

The model differs from traditional BOINC-style distributed computation in one crucial respect: the video being processed is human subjects data, and governance cannot be separated from computation. Masking@Home therefore integrates consent metadata as a first-class object: video owners specify, at the time of submission, which de-identification level is required, which communities may access which representation of the media, and under what archival licence the output may be deposited. The platform enforces these manifests throughout the processing pipeline; no output reaches the shared corpus without a provenance-complete consent record.

The potential scale is substantial. If even a fraction of the EAF files currently held in lab storage or informal archives were processed through Masking@Home and deposited in open repositories, the multimodal research community would have access to a corpus orders of magnitude larger than what is currently available. This corpus would, in turn, enable the AI pre-annotation models of Layer 2 to be trained on genuinely diverse data—not just the carefully curated and typically well-resourced corpora that currently dominate training sets—producing models that work across the full range of languages and interaction contexts that ELAN users have documented.

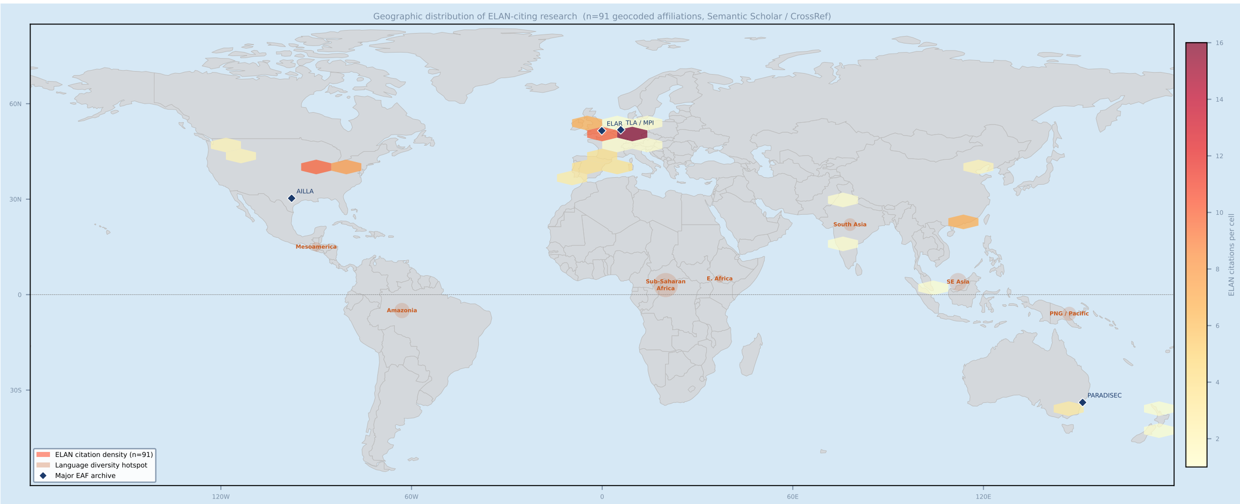

Figure 3: The geography of multimodal language research. Orange circles mark language diversity hotspots—regions of highest linguistic density and documentation urgency; circle area scales approximately with absolute speaker community count. Blue dots mark major multimodal research institutions; open circles indicate significant EAF archive holdings. The EU privacy regulation overlay (GDPR) is visible across Western Europe. The asymmetry between infrastructure concentration and diversity distribution motivates both the open stack and the Masking@Home initiative.

The Open Annotation Stack

We propose that the field's needs are best met not by improving ELAN but by building an *open annotation stack*—a layered, modular infrastructure in which each layer has a defined interface and can be swapped or extended independently. Figure 2 sketches the architecture. [6]

[6] The term *stack* is deliberate. Like the web stack (browser / HTTP / server / database), an annotation stack should be composable, with any compliant implementation substitutable at each layer without disrupting the others.

Layer 1: Data and Privacy. This layer handles ingestion, anonymization, consent metadata, and access control. It takes raw media as input and produces a *privacy-annotated media object*: the original alongside one or more derived representations (face-blurred, voice-altered, skeletal-only), together with a consent manifest specifying what may be shared with whom and under what conditions. Tools such as MaskingOPS (Owoyele 2025) are natural candidates for this layer. No raw media passes Layer 1 without explicit consent metadata; this structural constraint enforces privacy-by-design rather than privacy-by-convention.

Layer 2: Annotation Platform. This layer covers the functionality ELAN currently provides—time-aligned tier-based annotation, media playback, tier templates, inter-annotator agreement—but is designed as a *platform* rather than an application. It exposes an open API through which AI tools can pre-populate tiers, external validators can flag inconsistencies, and collaboration services can synchronize annotation state across institutions. Existing tools, including ELAN itself, could be adapted to conform to the layer's interface specification rather than being rebuilt from scratch; the goal is substitutability, not replacement.

Layer 3: Analysis and Export. This layer handles downstream consumption: corpus search, statistical export, archival deposit, and model training. It receives structured annotation objects from Layer 2 and produces outputs appropriate to each consumer. Critically, it is the natural home for feedback loops: AI models trained on corrected annotations can return improved pre-annotation to Layer 2, creating a virtuous cycle in which the cost of annotation decreases as corpus size grows.

Collaboration as First-Class Infrastructure

The three-layer stack is not only a technical architecture but a collaboration architecture. Current annotation practice is largely solitary: one researcher, one corpus, one ELAN session. The rare exceptions—inter-rater reliability studies involving two coders—are managed through manual file exchange and tool-dependent export routines that were not designed for synchronous multi-party work.

At Layer 2, collaboration requires a data model for annotation that tracks authorship, revision history, and conflict state. The analogy is version control for code: just as `git` allows multiple developers to work concurrently on a shared codebase with a principled merge strategy, an annotation platform should allow multiple annotators to work on the same corpus segment concurrently, with explicit protocols for resolving disagreements. Inter-annotator agreement is a validity criterion in every corpus-based field; the tools should make it a structural property of the workflow rather than a post-hoc calculation on exported files.

Role differentiation is equally important. A realistic annotation workflow involves at least three distinct roles: the field annotator who produces a first pass (often AI-assisted), the domain expert who adjudicates ambiguous cases, and the corpus manager who oversees quality and access control. These roles require different permissions, different views of the data, and different responsibilities for the final annotation state. Layer 2 should encode this role hierarchy explicitly. [7]

[7] Role-based annotation models are not new in the annotation systems literature (Stede 2012), but existing implementations treat roles as post-hoc access controls rather than as part of the data model. The distinction matters: if the role is part of the model, the audit trail is lossless.

Concretely, this implies that Layer 2 exposes not a single annotation state but a branched history: AI output, annotator corrections, adjudicated resolutions, and a final canonical tier—all preserved and attributable. This history is the audit trail that institutional ethics requirements increasingly demand, and it is the training signal that allows Layer 3 models to learn from human correction decisions rather than from final annotations in which the correction rationale has been erased.

Global Reach and Language Equity

Multimodal language research is not geographically neutral. The languages best documented—with richest annotated corpora, most developed computational tools, and most accessible archives—are predominantly the official languages of well-resourced states. The world's approximately 7,000 languages are distributed very differently: most are spoken by relatively small communities across Sub-Saharan Africa, South and Southeast Asia, Oceania, and the Americas (Eberhard et al. 2023). These are precisely the languages for which video-based documentation is most urgent—many lack writing systems, and gesture, prosody, and embodied practice are primary carriers of meaning.

The infrastructure barrier in these contexts is acute. Field linguists conducting documentation in Papua New Guinea, Cameroon, or the Amazon Basin bring institutional affiliations that provide ELAN licenses but not the computational resources or connectivity for AI-assisted pipelines. Annotation is done locally; corpora are uploaded to European or North American archives months later; community members—speakers of the documented language—have no formal role in the annotation workflow. The result is a knowledge production asymmetry that mirrors the historical dynamics of extractive fieldwork: communities provide the data; institutions far away hold the tools and the archives.

An open annotation stack changes this calculus structurally. When the stack's layers are accessible as open, cloud-compatible services, a community member with a mobile device and a reliable connection can participate in the annotation of their own language materials. Layer 1's privacy controls mean that consent is managed within the infrastructure rather than through paperwork separated from the data. Layer 2's role hierarchy means that community annotators and institutional researchers share the same corpus through a structured, attributable workflow. Layer 3's export layer can produce materials appropriate for community language programs as well as academic archives.

Figure 3 maps the current landscape. The concentration of multimodal research infrastructure in Western Europe and North America is visible against the distribution of language diversity hotspots—regions where documentation need is highest and tool access is most constrained. The open stack's value is not only academic productivity; it is the realignment of research infrastructure with the geography of need.

Community Governance

A stack is only as good as its governance. The EAF format's longevity demonstrates that community adoption of a shared standard is achievable; the fragmentation of tooling around EAF demonstrates that format alone is insufficient. The open annotation stack requires a governance body—modeled on initiatives such as CLARIN (Hinrichs et al. 2010), the W3C Web Annotation Working Group, or the OpenStreetMap Foundation—that maintains interface specifications, certifies compliant implementations, and coordinates the developer community.

Critically, governance must include not only technical and institutional representatives but also language community stakeholders. The communities whose materials constitute much of the existing EAF heritage corpus have historically been positioned as data sources rather than as co-stewards of the infrastructure built around their languages. A governance model that includes community language programs, indigenous data sovereignty advocates, and field linguists working in non-European contexts is not merely more equitable; it is more epistemically robust, because it surfaces requirements—about consent, about access, about representation—that institution-only governance bodies tend to overlook. [8]

[8] The CARE Principles for Indigenous Data Governance (Carroll et al. 2020) provide a framework for community-centred data stewardship that is directly applicable to the annotation stack governance model.

Certification requires defining what compliance means at each layer: which annotation primitives are mandatory, which are optional, how versioning is handled, how consent manifests are validated, and how implementations signal capability to downstream consumers. These questions do not have obvious answers.

Open Research Questions

Building the stack surfaces genuine research problems that cannot be resolved by engineering alone:

- **Provenance representation.** How should anonymization transforms be encoded in annotation metadata so that downstream consumers know which media version a tier references and the full provenance chain is machine-readable?

- **Human-AI annotation interfaces.** What interaction patterns enable efficient review of AI pre-annotations at scale, and how does cognitive load shift when annotators move between correction and first-pass creation modes?

- **Agreement with AI-generated tiers.** How should inter-annotator agreement be computed when tiers are partially machine-generated, and to what degree should model confidence propagate into agreement statistics?

- **Consent and archive interoperability.** How do consent manifests interact with long-term deposit requirements across jurisdictions, and what mechanisms enforce access controls as archives transfer between custodians?
- **Community data sovereignty.** How can the stack accommodate indigenous data governance frameworks (Carroll et al. 2020) that require community control over materials, and what incentive structures make Masking@Home sustainable as a community-driven initiative (Borgman 2012)?

These questions require sustained collaboration across computational linguistics, human–computer interaction, research data management, archival science, and information ethics. They are the kind of cross-disciplinary infrastructure problems that have historically been underserved by both domain grant programmes (which fund research *with* tools, not research *on* tools) and engineering grants (which fund building, not the governance and design research that precedes it). They constitute a research programme in their own right.

Conclusion

ELAN is not broken. It works well within the constraints for which it was designed. Those constraints no longer match what the field needs. The path forward is to build the infrastructure beneath and around it: a stack that treats privacy, AI integration, interoperability, and collaboration as design primitives rather than optional extensions.

The EAF heritage corpus is an underappreciated asset. Two decades of annotation labor is encoded in EAF files distributed across labs, laptops, and institutional servers around the world. Much of it cannot be shared in its current form. The Masking@Home initiative offers a community-driven mechanism to unlock it: apply privacy-preserving anonymization at scale, return de-identified media to open archives, and allow the field to build on the collective investment it has already made.

The open annotation stack is not a single tool and not a single institution's project. It is community infrastructure—the kind that funding bodies are increasingly willing to treat as a legitimate and high-impact research target. Investment in interface specifications, reference implementations, governance structures that include language communities, and the empirical research questions that underpin these components would position the multimodal research community to absorb the next decade of AI capability and regulatory change without accumulating the technical and ethical debt that currently makes every lab's workflow its own private island.

The field already has most of the data. What it lacks is the infrastructure to share it.

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